S I S S O 的 安 装 与 使 用

N A N J I N G U N I V E R S I T Y

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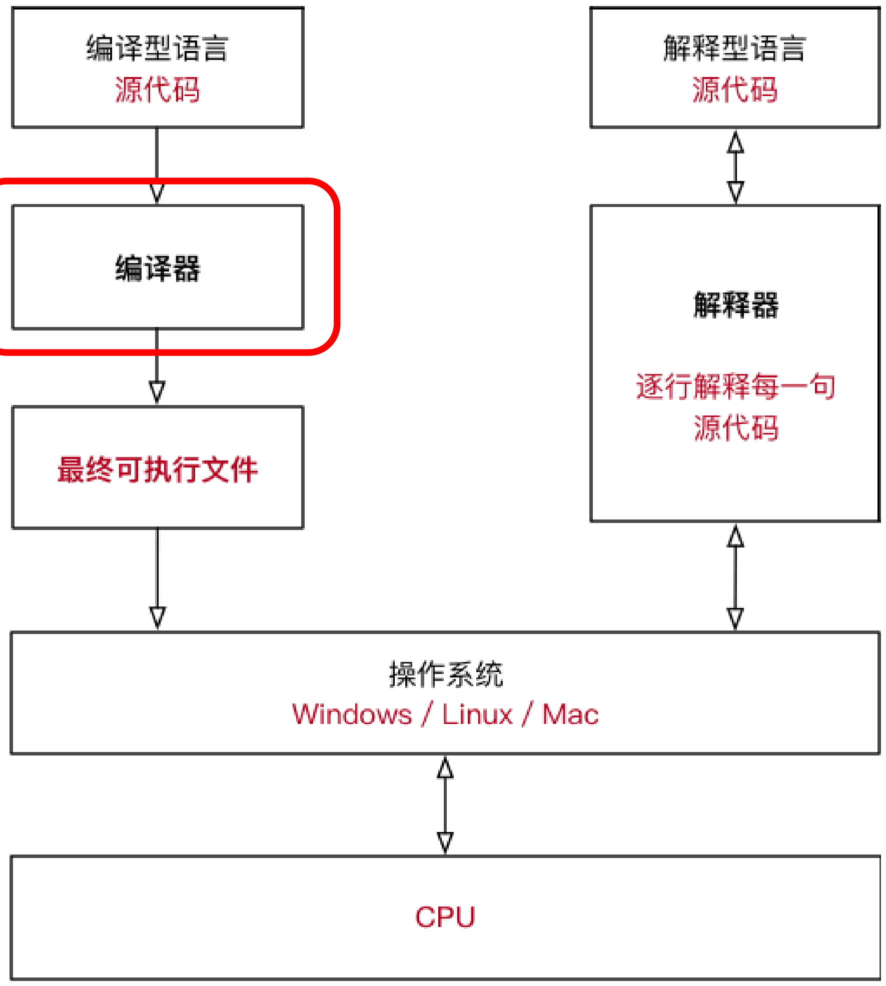
编译型语言与解释性语言的区别

C语言
C++
Fortran



Python
JavaScript
MATLAB
SHELL

SISSO源代码已有，需要安装编译器

[1.2 编译型语言和解释型语言的区别 - 赛兔子 - 博客园]

[编译型语言和解释型语言的区别_如何保存解释型语言结果-CSDN博客]

SISSO推荐使用Intel MPI 编译器

2. Installation

A FORTRAN MPI compiler is required to compile the SISSO parallel program. Below are two options for compiling the program using an Intel MPI compiler (other compilers may work as well but the Intel compilers are recommended). Download and unzip the package in Linux operating systems, and then go to the folder 'src' and do:

1) mpiifort -fp-model precise var_global.f90 libsisso.f90 DI.f90 FC.f90 SISSO.f90 -o ~/bin/SISSO

or

2) mpiifort -O2 var_global.f90 libsisso.f90 DI.f90 FC.f90 SISSO.f90 -o ~/bin/SISSO

The option (1) enables better accuracy and run-to-run reproducibility of floating-point calculations; option (2) is about 2X faster than (1) but tiny run-to-run variations may happen between processors of different types, e.g. Intel and AMD.

If 'mpi' related errors present during the compilation, try opening the file 'var_global.f90' and replace the line "use mpi" with "include 'mpif.h'". However, " use mpi " is strongly encouraged (see https://www.mpi-forum.org/docs/mpi-3.1/mpi31-report/node411.htm).

[什么是MPI-CSDN博客]

[并行计算之MPI（一）-CSDN博客]

Fortran MPI 编译器mpiifort 安装

- Intel mpiifort编译器集成在[Intel oneAPI HPC Toolkit]中
- 文件名
  - l_HPCKit_p_2024.2.1.79_offline.sh
- 安装命令
  - sudo chmod +x l_HPCKit_p_2024.2.1.79_offline.sh （给予文件运行权限）
  - sh l_HPCKit_p_2024.2.1.79_offline.sh
- 选择Intel MPI Library 以及 Intel Fortran Compiler安装
- 加载intel oneapi环境
  - source ~/intel/oneapi/setvars.sh --force
  - 可加入~/.bashrc

GCC安装

- GCC（GNU 编译器集合）是一个开源的编译器工具集，支持多种编程语言的编译，包括 C、C++、Fortran、Ada 等。GCC 是 Linux 和许多其他操作系统中最常用的编译器之一，广泛应用于软件开发和系统编程中。

- 安装指令
  - sudo apt install gcc

- 服务器上离线安装gcc
  - [linux下离线安装gcc详细教程_linux离线安装gcc-CSDN博客]()

确定安装完成

- gcc -v

编译SISSO软件

- 安装rar压缩包解压文件
  - sudo apt install unrar
  - unrar x SISSO.rar

- 进入SISSO/src文件夹下 (在oneapi环境下)
  - 运行
    - 创建文件夹： mkdir ~/bin
    - mpiifort -fp-model precise var_global.f90 libsisso.f90 DI.f90 FC.f90 SISSO.f90 -o ~/bin/SISSO

  - 完成SISSO编译

将SISSO加入系统环境变量

- 将路径加入系统环境变量后可在Linux系统中任意位置执行该路径下的可执行文件
- vim ~/.bashrc
  - export PATH=~/bin:$PATH
- source ~/.bashrc
- 在任意位置输入SISSO即可运行程序

SISSO的使用

- train.dat文件结构
- SISSO.in 配置文件
- 详见SISSO_Guide.pdf

SISSO运行指令

- 单核运行
  - SISSO > log

- 多核运行
  - mpirun -np [number_of_cores] SISSO > log
  - [number_of_cores] = 2 | 4 | 24 | 48 ……

SISSO作业

- SI.zip
  - 取自文献的数据集　　unzip SI.zip

materials	property	Si/Al	PLD	Nd	H2	X	R_{DLS}	G	P	AN	R	IE
sample1	-3.5032	95	6.36	1	16.11	1.36	0.002	3	4	21	144	633.1
sample2	-1.2317	95	6.36	2	18.6	1.54	0.002	4	4	22	132	658.8
sample3	0.69593	95	6.36	3	20.8	1.63	0.002	5	4	23	122	650.9
sample4	0.85579	95	6.36	5	20	1.66	0.002	6	4	24	118	652.9
sample5	1.00094	95	6.36	5	14.64	1.55	0.002	7	4	25	117	717.3
sample6	0.81326	95	6.36	6	13.8	1.83	0.002	8	4	26	117	762.5
sample7	0.43149	95	6.36	7	16.19	1.88	0.002	9	4	27	116	760.4
sample8	-0.45364	95	6.36	8	17.2	1.91	0.002	10	4	28	115	737.1
sample9	-1.13561	95	6.36	10	13.14	1.9	0.002	11	4	29	117	745.5
sample10	-1.25336	95	6.36	10	7.38	1.65	0.002	12	4	30	125	906.4
sample11	-3.81395	95	6.36	1	17.5	1.22	0.002	3	4	39	162	600
sample12	-1.89309	95	6.36	2	21	1.33	0.002	4	5	40	145	640.1
sample13	2.45907	95	6.36	4	26.9	1.6	0.002	5	5	41	134	652.1
sample14	2.9795	95	6.36	5	36	2.16	0.002	6	5	42	130	684.3
sample15	1.7391	95	6.36	7	25.5	2.2	0.002	8	5	44	125	710.2
sample16	1.01484	95	6.36	8	21.8	2.28	0.002	9	5	45	125	719.7
sample17	0.31358	95	6.36	10	16.74	2.2	0.002	10	5	46	128	804.4
sample18	-1.36593	95	6.36	10	11.3	1.93	0.002	11	5	47	134	731
sample19	-1.64629	95	6.36	10	6.07	1.69	0.002	12	5	48	148	867.8
sample20	1.07586	95	6.36	3	36	1.5	0.002	5	6	73	134	761
sample21	3.28787	95	6.36	4	35	2.36	0.002	6	6	74	130	770
sample22	4.41707	95	6.36	5	33.05	1.9	0.002	7	6	75	128	760
sample23	3.75011	95	6.36	6	29.29	2.2	0.002	8	6	76	126	840
sample24	2.81585	95	6.36	7	26.36	2.2	0.002	9	6	77	127	880
sample25	2.0665	95	6.36	9	19.66	2.28	0.002	10	6	78	130	870
sample26	-1.61343	95	6.36	0	4.77	1.8	0.002	14	6	82	147	715.6
sample27	-7.17629	47	6.36	1	16.11	1.36	0.002	3	4	21	144	633.1

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Nitrogen reduction reaction energy and pathways in metal-zeolites: deep learning and explainable machine learning with local acidity and hydrogen bonding features†

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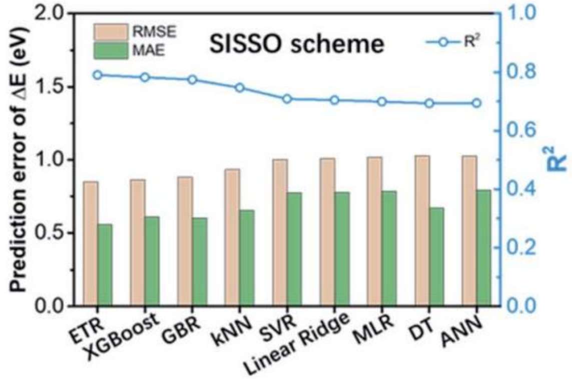
3 SISSO features

$$SISSO1 = e^{\frac{N_H}{N_N}} + e^{PLD*\Delta Q_{CT}}$$

$$SISSO2 = \frac{d_{MN}^2 / e^{\Delta Q_{CT}}}{e^{N_N} + e^{N_H}}$$

$$SISSO3 = \frac{lgN_N}{N_N} N_{HB} + \frac{lgPLD}{e^{LA}}$$



e.g., NH_2* (Ti-zeolite)

ΔE^{SISSO} = -0.1272·SISSO1 + 4.3625·SISSO2 + 1.5625·SISSO3 - 4.8158
= (-1.0537 + 4.7030 + 0.9130 – 4.8158) eV
= -3.47 eV (SISSO)

ΔE^{DFT} = -3.71 eV (DFT)

欢迎提问！